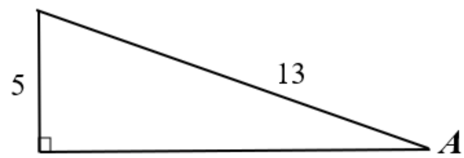
DPEN022: Enabling Mathematics B
Solutions to Class Questions Week 1

<u>Week 1A</u>

Trigonometry - Right-Angle Triangles

1. (i) $\sin \varphi = \dfrac{4}{5}$ (ii) $\tan \theta = \dfrac{3}{4}$ (iii) $\operatorname{cosec} \varphi = \dfrac{5}{4}$ (iv) $\cot \theta = \dfrac{4}{3}$

2.



Using Pythagoras Theorem, the last side is calculated.

$$\begin{aligned} x &= \sqrt{13^2 - 5^2} \\ &= 12 \end{aligned}$$

(i) $\cos A = \dfrac{12}{13}$ (ii) $\tan A = \dfrac{5}{12}$ (iii) $\operatorname{cosec} A = \dfrac{1}{\sin A} = \dfrac{13}{5}$

3. (i) $\sin B = \dfrac{\sqrt{7}}{4}$ (ii) $\tan B = \dfrac{\sqrt{7}}{3}$ (iii) $\sec B = \dfrac{1}{\cos B} = \dfrac{4}{3}$

4. (i) $\sin C = \dfrac{3}{\sqrt{10}}$ (ii) $\cos C = \dfrac{1}{\sqrt{10}}$ (iii) $\tan C = 3$

5. (i) $\sin 39^\circ 25' = 0.635$ (ii) $\cos 45^\circ 50' = 0.697$ (iii) $\cot 68.5^\circ = 0.394$

6. (i) $\sin\theta = 0.298$

$$\begin{aligned} \theta &= \sin^{-1} 0.298 \\ &= 17^\circ 20' \end{aligned}$$

(ii) $\tan\theta = 0.683$

$$\begin{aligned} \theta &= \tan^{-1} 0.683 \\ &= 34^\circ 20' \end{aligned}$$

(iii) $\sin\theta = \dfrac{1}{1.827}$

$$\theta = \sin^{-1}\left(\frac{1}{1.827}\right) = 33^\circ 11'$$

7. (i) $\sin 31^\circ 43' = \dfrac{x}{12}$

$$x = 12 \times \sin 31^\circ 43'$$
$$x = 6.3$$

(ii) $\sin 51^\circ 14' = \dfrac{8.9}{x}$

$$x = \frac{8.9}{\sin 51^\circ 14'}$$
$$x = 11.4$$

(iii) $\tan 45^\circ 30' = \dfrac{x}{3.8}$

$$x = 3.8 \tan 45^\circ 30'$$
$$x = 3.9cm$$

8. (i) $\tan\beta = \dfrac{16.7}{5.8}$

$$\begin{aligned} \beta &= \tan^{-1}\left(\frac{16.7}{5.8}\right) \\ &= 71^\circ \end{aligned}$$

(ii) $\sin\alpha = \dfrac{16.7}{29.4}$

$$\begin{aligned} \alpha &= \sin^{-1}\left(\frac{16.7}{29.4}\right) \\ &= 35^\circ \end{aligned}$$

(iii) $\cos\alpha = \dfrac{15.7}{43.9}$

$$\begin{aligned} \alpha &= \cos^{-1}\left(\frac{15.7}{43.9}\right) \\ &= 69^\circ \end{aligned}$$

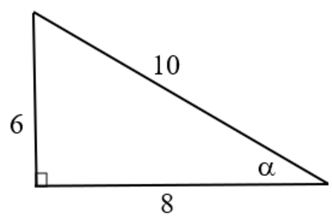
9. (i)

$$\begin{aligned} x &= \sqrt{10^2 - 6^2} \\ &= 8 \end{aligned}$$

The smallest angle is opposite the smallest side.

$$\sin\alpha = \frac{6}{10}$$

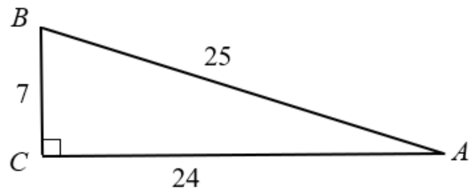
$$\begin{aligned} \alpha &= \sin^{-1}(0.6) \\ &= 36.9^\circ \end{aligned}$$



(ii) $\sin A = \dfrac{7}{25}$

$A = \sin^{-1}\left(\dfrac{7}{25}\right)$

$= 16.3°$



Since angles A and B are complementary, angle A = 90 – 16.3 = 73.7°.

Alternatively, $\sin B = \dfrac{24}{25}$

$B = \sin^{-1}\left(\dfrac{24}{25}\right)$

$= 73.7°$

Trigonometry - Complementary Angles

1. (i)
$$\sin x^\circ = \cos 50^\circ$$
$$x = 40^\circ$$

(ii)
$$\sin 80° = \cos(90 - 2x)°$$
$$\cos 10° = \cos(90 - 2x)°$$
$$10 = 90 - 2x$$
$$2x = 80$$
$$x = 40$$

(iii)
$$\tan 80 = \cot(2x)^\circ$$
$$\cot 10 = \cot(2x)^\circ$$
$$2x = 10$$
$$x = 5$$

(iv)
$$\cos x = \sin(x + 20)^\circ$$
$$\cos x = \sin(90 - x)^\circ$$
$$\therefore x + 20 = 90 - x$$
$$2x = 70$$
$$x = 35^\circ$$

2. (i)
$$\sin x + \cos(90^\circ - x) - 2\sin x$$
$$= \sin x + \sin x - 2\sin x$$
$$= 0$$

(ii)
$$\sec x - \cos ec(90° - x)$$
$$= \sec x - \sec x$$
$$= 0$$

3. (i)
$$\frac{\sin 50^\circ}{\cos 40^\circ} = \frac{\sin 50^\circ}{\sin 50^\circ}$$
$$= 1$$

(ii)
$$\frac{\tan 35^\circ}{\cot 55^\circ} = \frac{\tan 35^\circ}{\tan 35^\circ}$$
$$= 1$$

(iii)
$$\sec 20^\circ - \cos ec 70^\circ$$
$$= \sec 20^\circ - \sec 20^\circ$$
$$= 0$$

Trigonometry - Exact Trigonometric Ratios 30°, 45°, 60°

1. (i)
$$\sin 60° + \cos 60° = \frac{\sqrt{3}}{2} + \frac{1}{2}$$
$$= \frac{\sqrt{3} + 1}{2}$$

(ii)
$$\cos^2 45° + \sin^2 45° = \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2$$
$$= \frac{1}{2} + \frac{1}{2} = 1$$

(iii) $2\sec 60° = 2\times\frac{1}{\cos 60°}$

$= 2\times\frac{1}{\frac{1}{2}}$

$= 2\times 2$

$= 4$

(iv) $\operatorname{cosec} 45° = \frac{1}{\sin 45°}$

$= \frac{1}{\frac{1}{\sqrt{2}}}$

$= \sqrt{2}$

2. (i)

$$\sin\theta = \frac{1}{\sqrt{2}}$$

$$\theta = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$= 45°$$

(ii)

$$\cos\theta = \frac{1}{\sqrt{2}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$= 45°$$

(iii)

$$\tan\theta = 1$$

$$\theta = \tan^{-1}(1)$$

$$= 45°$$

(iv)

$$\cot\theta = \frac{1}{\tan\theta} \therefore \tan\theta = \frac{1}{\frac{1}{\sqrt{3}}} = \sqrt{3}$$

$$\theta = 60°$$

(v) $\sec\theta = \frac{1}{\cos\theta} \therefore \cos\theta = -\frac{1}{2}$

$$\cos\theta = \frac{1}{2}$$

$$\theta = \cos^{-1}\left(\frac{1}{2}\right)$$

$$= 60°$$

(vi) $\operatorname{cosec}\theta = \frac{1}{\sin\theta} \therefore \sin\theta = \frac{1}{\sqrt{2}}$

$$\sin\theta = \frac{1}{\sqrt{2}}$$

$$\theta = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$= 45°$$

Week 1B

Trigonometry - Angles of Any Magnitude

1. (i) 110° 2nd (ii) 210° 3rd (iii) 39° 1st
   (iv) 308° 4th (v) 281° 4th (vi) 97° 2nd
   (vii) 456° 2nd (viii) -25° 4th (ix) -135° 3rd

2.
   (i) sin 170° (positive) (ii) cos 280° (positive) (iii) tan 60° (positive)
   (iv) tan 125° (negative) (v) sin 56° (positive) (vi) cos 226° (negative)
   (vii) cot 13° (positive) (viii) cosec 320° (negative) (ix) sec 165° (negative)
   (x) cos (-45°) (positive) (xi) sin 370° (positive) (xii) sin (-125°) (negative)

3.

Angle θ	Quadrant	$\sin\theta$	$\cos\theta$	$\tan\theta$
120°	2	+	–	–
300°	4	–	+	–
240°	3	–	–	+
460°	2	+	–	–

4. (i) $\sin\theta > 0,\ \cos\theta < 0$ 2nd (ii) $\tan\theta > 0,\ \sin\theta < 0$ 3rd
   (iii) $\cos\theta > 0,\ \sin\theta > 0$ 1st (iv) $\tan\theta < 0,\ \cos\theta > 0$ 4th

Trigonometry - Exact Trigonometric Ratios 0°, 90°, 180°, 270° 360°

1.

Angle θ	Coordinates	$\sin\theta$	$\cos\theta$
0 °	(1, 0)	0	1
90°	(0,1)	1	0
180°	(-1,0)	0	-1
270°	(0,-1)	-1	0
360 °	(1,0)	0	1

2.
   (i) 4 cos 0° = 4 x 1 = 4
   (ii) tan 180° + tan 0° = 0 + 0 = 0
   (iii) sin 90° + sin 180° = 1 + 0 = 1
   (iv) cos 540° = cos (540-360) = cos 180° = -1
   (v) cos 180° sin 270° = -1 x -1 = 1
   (vi) 2 cos 90° sin 90° = 2 x 0 x 1 = 0
   (vii) $\sin^2 270 + \cos^2 270 = (-1)^2 + (0)^2$
   $= 1 + 0 = 1$
   (viii) cosec 180° = $\frac{1}{\sin 180°} = \frac{1}{0} = undefined$
   (ix) 3 sec 0°- tan 180° = $\frac{3}{\cos 0} - 0 = 3 - 0 = 3$

Trigonometry – Angles of any Magnitude

1. (i) $$\cos 150° = \cos(180° - 150°) \\ = -\cos 30° \\ = \frac{-\sqrt{3}}{2}$$

(ii) $$\tan 150° = \tan(180° - 150°) \\ = -\tan 30° \\ = \frac{-1}{\sqrt{3}}$$

(iii) $$\sin 240° = -\sin(240° - 180°) \\ = -\sin(60°) \\ = \frac{-\sqrt{3}}{2}$$

(iv) $$\cos 315° = \cos(360° - 45°) \\ = \cos 45° \\ = \frac{1}{\sqrt{2}}$$

(v) $$\sin(-120°) = \sin(360° - 120°) = \sin 240° \\ = \sin(240° - 180°) \\ = -\sin(60°) \\ = -\frac{\sqrt{3}}{2}$$

(vi) $$\tan 570° = \tan(570° - 360°) \\ = \tan 210° \\ = \tan(210° - 180°) \\ = \tan(30°) \\ = \frac{1}{\sqrt{3}}$$

(vii) $$\tan -45° = -\tan 45° \\ = -1$$

(viii) $$\cos 210° = -\cos 30 \\ = \frac{-\sqrt{3}}{2}$$

(ix) $$\sin -300° = \sin 60° \\ = \frac{\sqrt{3}}{2}$$

(x) $$\cos ec 300° = \frac{1}{-\sin 60} \\ = \frac{1}{\frac{-\sqrt{3}}{2}} = -\frac{2}{\sqrt{3}}$$

(xi) $$\cot 135° = \frac{1}{-\tan 45} \\ = \frac{1}{-1} = -1$$

(xii) $$\sec 225° = \frac{1}{-\cos 45} \\ = \frac{1}{\frac{-1}{\sqrt{2}}} = -\sqrt{2}$$

Trigonometry - Solving Trigonometric Equations in Degrees

1. (i) $\sin \theta$ is positive therefore θ lies in quadrant 1 and quadrant 2

Reference angle:

$$\sin \theta_r = \frac{1}{\sqrt{2}}$$

$$\theta_r = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \\ = 45°$$

Therefore $\theta = 45°$ and $180° - 45°$

$$\theta = 45°, 135°$$

(ii) $\cos\theta$ is positive therefore θ lies in quadrant 1 and quadrant 4

Reference angle:

$$\cos\,\theta_r = \frac{1}{\sqrt{2}}$$

$$\theta_r = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$$
$$= 45°$$

Therefore $\theta = 45^\circ$ and $360° - 45^\circ$

$\theta = 45°, 315°$

(iii) $\tan\theta$ is negative therefore θ lies in quadrant 2 and quadrant 4

Reference angle:

$$\theta_r = \tan^{-1}(1)$$
$$= 45°$$

Therefore $\theta = 180° - 45°$ and $360° - 45°$

$\theta = 135°, 315°$

(iv) $\sin\theta$ is negative therefore θ lies in quadrant 3 and quadrant 4

Reference angle:

$$\theta_r = \sin^{-1}\left(\frac{1}{2}\right)$$
$$= 30°$$

Therefore $\theta = 180° + 30°$ and $360° - 30°$

$\theta = 210°, 330°$

(v) $\tan\theta$ is positive therefore θ lies in quadrant 1 and quadrant 3

Reference angle:

$$\theta_r = \tan^{-1}\left(\sqrt{3}\right)$$
$$= 60°$$

Therefore $\theta = 60^\circ$ and $\theta = 180^\circ + 60^\circ$

$\theta = 60^\circ, 240^\circ$

(vi) $\cot\theta = \dfrac{1}{\sqrt{3}} \rightarrow \dfrac{1}{\tan\theta} = \dfrac{1}{\sqrt{3}} \rightarrow \tan\theta = \sqrt{3}$

$\tan\theta$ is positive therefore θ lies in quadrant 1 and quadrant 3.

Reference angle:

$$\theta_r = \tan^{-1}\left(\sqrt{3}\right)$$
$$= 60°$$

Therefore $\theta = 60^\circ$ and $\theta = 180^\circ + 60^\circ$

$\theta = 60°, 240°$

(vii) $\sec\theta = \dfrac{1}{\cos\theta} \therefore \cos\theta = -\dfrac{1}{2}$

$\cos\theta$ is negative therefore θ lies in quadrant 2 and quadrant 3

Reference angle:

$$\theta_r = \cos^{-1}\left(\frac{1}{2}\right)$$
$$= 60°$$

Therefore $\theta = 180^\circ - 60^\circ$ and $\theta = 180^\circ + 60^\circ$

$\theta = 120^\circ, 240^\circ$

(viii) $\operatorname{cosec}\theta = \dfrac{1}{\sin\theta} \therefore \sin\theta = \dfrac{1}{\sqrt{2}}$

$\sin\theta$ is positive therefore θ lies in quadrant 1 and quadrant 2

Reference angle:

$$\theta_r = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

Therefore $\theta = 45^\circ$ and $180^\circ - 45^\circ$

$\theta = 45^\circ, 135^\circ$

(ix) $2\cos\theta = -1$

$\cos\theta = \dfrac{-1}{2}$

$\cos\theta$ is negative therefore θ lies in quadrant 2 and quadrant 3

Rederence angle:

$$\theta_r = \cos^{-1}\left(\frac{1}{2}\right) = 60^\circ$$

Therefore $\theta = 180^\circ - 60^\circ, 180^\circ + 60^\circ$

$= 120^\circ, 240^\circ$

(x) $\tan^2\theta = 1$

$\tan\theta = \pm\sqrt{1} = \pm 1$

$\theta_r = \tan^{-1}(1)$

$= 45^\circ$

therefore θ lies in Q_1, Q_2, Q_3 and Q_4

$\theta = 45^\circ, 180^\circ - 45^\circ, 180^\circ + 45^\circ, 360^\circ - 45^\circ$

$= 45^\circ, 135^\circ, 225^\circ, 315^\circ$

(xi) $\sqrt{3}\tan\theta = 1$

$\tan\theta = \dfrac{1}{\sqrt{3}}$

$\tan\theta$ is positive therefore θ lies in quadrant 1and quadrant 3

$\theta_r = \tan^{-1}\left(\dfrac{1}{\sqrt{3}}\right)$

$= 30^\circ$

$\theta = 30^\circ, 180^\circ + 30^\circ$

$= 30^\circ, 210^\circ$

(xii) $\cos^2\theta = \dfrac{1}{4} \Rightarrow \cos\theta = \pm\dfrac{1}{2}$

$\cos\theta$ is positive and negative therefore θ lies in Q_1, Q_2, Q_3 and Q_4

$\theta = 60^\circ, 180^\circ \pm 60^\circ, 360^\circ - 60^\circ$

$= 60^\circ, 120^\circ, 240^\circ, 300^\circ$

2\. (i) $\sin\theta$ is positive therefore θ lies in quadrant 1 and quadrant 2

Reference angle:

$\sin\theta_r = 0.4827$

$\theta_r = \sin^{-1}(0.4827)$

$= 28^\circ 52'$

Therefore $\theta = \theta_r$ and $180^\circ - \theta_r$

$\theta = 28^\circ 52'$ & $\theta = 180^\circ - 28^\circ 52' = 151^\circ 08'$

(ii) $\cos\theta$ is negative therefore θ lies in quadrant 2 and quadrant 3

Reference angle:

$$\cos\theta_r = 0.8731$$
$$\theta_r = \cos^{-1}(0.8731)$$
$$= 29°11'$$

Therefore $\theta = 180 - \theta_r$ and $180° + \theta_r$

$$\theta = 180° - 29°11' \quad \text{and} \quad \theta = 180° + 29°11'$$
$$= 150°49' \qquad\qquad = 209°11'$$

(iii) $\operatorname{cosec}\theta = \dfrac{1}{\sin\theta} \therefore \sin\theta = \dfrac{1}{3.6771}$

is positive therefore θ lies in quadrant 1 and quadrant 2

Reference angle:

$$\sin\theta_r = \frac{1}{3.6771}$$
$$\theta_r = \sin^{-1}\left(\frac{1}{3.6771}\right)$$
$$= 15°47'$$

Therefore $\theta = \theta_r$ and $180° - \theta_r$

$$\theta = 15°47' \quad \& \quad \theta = 180° - 15°47' = 164°13'$$

Week 1C

Trigonometry - Radian Measure

1. (i) $120° = 120° \cdot \dfrac{\pi}{180°} = \dfrac{2\pi}{3}\, rad$

   (ii) $18° = 18° \cdot \dfrac{\pi}{180°} = \dfrac{\pi}{10}\, rad$

   (iii) $-225° = -225° \cdot \dfrac{\pi}{180°} = -\dfrac{5\pi}{4}\, rad$

2. (i) $\dfrac{17\pi}{12} = \dfrac{17\pi}{12} \cdot \dfrac{180°}{\pi} = 255°$

   (ii) $1.2 = 1.2 \cdot \dfrac{180°}{\pi} = \dfrac{216}{\pi} = 68°45'$

   (iii) $-3.15 = -3.15 \cdot \dfrac{180°}{\pi} = -\dfrac{567}{\pi} = -180°29'$

3. (i) 1 (ii) 2 (iii) 3 (iv) 4
   (v) 2 (vi) 1 (vii) 2 (viii) 1

Trigonometry – Angles of any Magnitude in Radians

4.

(i) 0 (ii) -1 (iii) $\dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$ (iv) $\dfrac{1}{2}$

(v) $-\dfrac{1}{2}$ (vi) 1 (vii) 0 (viii) 1

(ix) -1 (x) $\dfrac{\sqrt{3}}{2}$ (xi) $\dfrac{1}{2}$ (xii) $-\dfrac{\sqrt{3}}{2}$

(xiii) $\dfrac{1}{2}$ (xiv) $-\dfrac{\sqrt{2}}{2}$ (xv) $\sqrt{3}$ (xvi) $-\dfrac{1}{2}$

Trigonometry - Solving Trigonometric Equations in Radians

1. (i) $\sin x = -\dfrac{1}{\sqrt{2}}$ Sin is negative in Q 3 and Q 4.

$x_R = \sin^{-1}\left(\dfrac{1}{\sqrt{2}}\right) = \dfrac{\pi}{4}$

$$x = \pi + \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$
$$= \frac{5\pi}{4}, \frac{7\pi}{4}$$

(ii) $\cos x = \dfrac{1}{\sqrt{2}}$ Cos is positive in Q 1 and Q 4.

$x_R = \cos^{-1}\left(\dfrac{1}{\sqrt{2}}\right) = \dfrac{\pi}{4}$

$$x = \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$
$$= \frac{\pi}{4}, \frac{7\pi}{4}$$

(iii) $\tan x = -1$ Tan is negative in Q 2 and Q 4.

$$x_R = \tan^{-1} 1 = \frac{\pi}{4}$$

$$x = \pi - \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$
$$= \frac{3\pi}{4}, \frac{7\pi}{4}$$

(iv) $\sin x = \frac{1}{2}$ Sin is positive in Q 1 and Q 2.

$$x_R = \sin^{-1} \frac{1}{2} = \frac{\pi}{6}$$

$$x = \frac{\pi}{6}, \pi - \frac{\pi}{6}$$
$$= \frac{\pi}{6}, \frac{5\pi}{6}$$

(v) $x = \frac{\pi}{3}, \frac{4\pi}{3}$

(vi) $x = \frac{\pi}{3}, \frac{4\pi}{3}$

(vii) $x = \frac{2\pi}{3}, \frac{4\pi}{3}$

(viii) $x = \frac{\pi}{4}, \frac{3\pi}{4}$

(ix) $x = \frac{2\pi}{3}, \frac{4\pi}{3}$

(x) $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

(xi) $x = \frac{\pi}{6}, \frac{7\pi}{6}$

(xii) $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

2. (i) $\sin^{-1}(0.4827) = 0.504, \pi - 0.504$

$$x = 0.504, 2.639 rad$$

(ii)

$$\operatorname{cosec} x = 3.6771$$
$$\sin x = \frac{1}{3.6771}$$
$$\sin^{-1}\left(\frac{1}{3.6771}\right) = 0.275, \pi - 0.275 rad$$
$$x = 0.275, 2.866 rad$$

Week 1D

Trigonometry - Graphs of Trigonometric Functions

1. (i) All real x $\{x \in \mathbb{R}\}$ (ii) $\{y \in \mathbb{R} : -1 \le y \le 1\}$ (iii) $2\pi, 1$ (iv) $\dfrac{2\pi}{n}, a$

2. (i) $y = \cos 2x$ (ii) $y = 2\cos x$ (iii) $y = \dfrac{1}{4}\cos 4x$ (iv) $y = -4\cos\dfrac{x}{4}$

   Amplitude = 1 Amplitude = 2 Amplitude = $\dfrac{1}{4}$ Amplitude = $|-4| = 4$

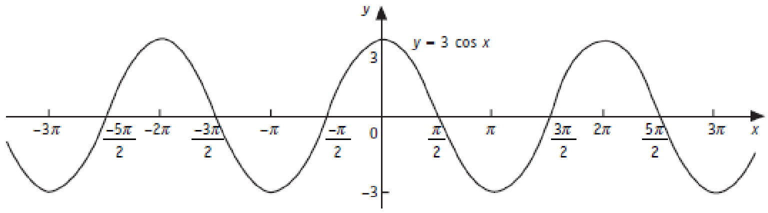
   Period = π Period = 2π Period = $\dfrac{\pi}{2}$ Period = 8π

   (v) $y = 3\cos 2x$ (vi) $y = 2 + 2\cos x$ (vii) $y = -0.5\cos 2\pi x$

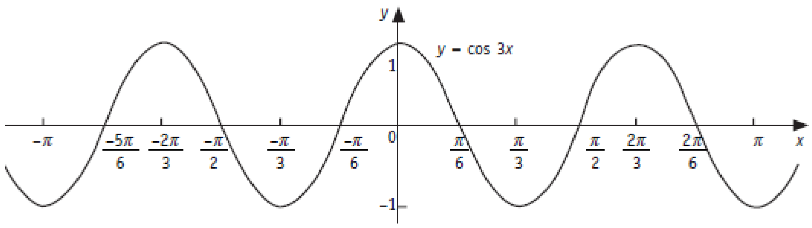
   Amplitude = 3 Amplitude = 2 Amplitude = 0.5

   Period = $\dfrac{2\pi}{b} = \dfrac{2\pi}{2} = \pi$ Period = 2π Period = $\dfrac{2\pi}{b} = \dfrac{2\pi}{2\pi} = 1$

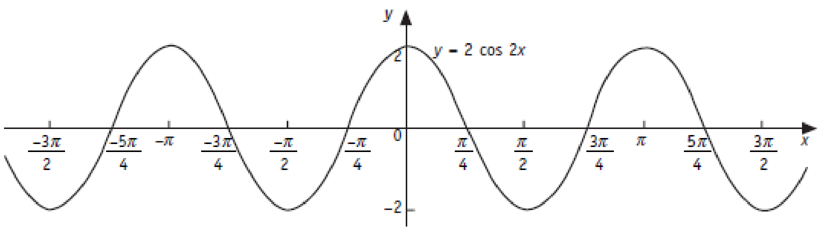
3. (i) $y = 3\cos x$



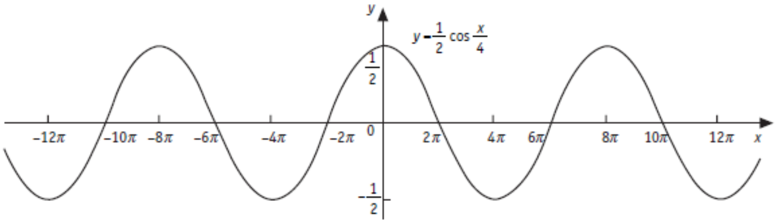
(ii) $y = \cos 3x$



(iii) $y = 2\cos 2x$



(iv) $y=\dfrac{1}{2}\cos\dfrac{x}{4}$



4. (i) All real x $\{x \in \mathbb{R}\}$ (ii) $\{y \in \mathbb{R} : -1 \le y \le 1\}$ (iii) $2\pi, 1$ (iv) $\dfrac{2\pi}{n}, a$

5. (i) $y=\sin 4x$ (ii) $y=4\sin x$ (iii) $y=\dfrac{1}{2}\sin 2x$ (iv) $y=-2\sin\dfrac{x}{4}$

Amplitude = 1   Amplitude = 4   Amplitude = $\dfrac{1}{2}$ Amplitude = $|-2|=2$

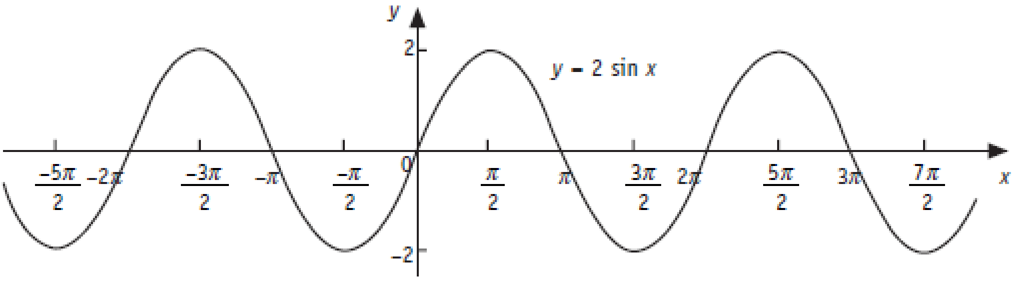
Period = $\dfrac{\pi}{2}$ Period = 2π Period = π Period = 8π

(v) $y=2\sin x$ (vi) $y=3\sin 4x$ (vii) $y=4\sin \pi x$

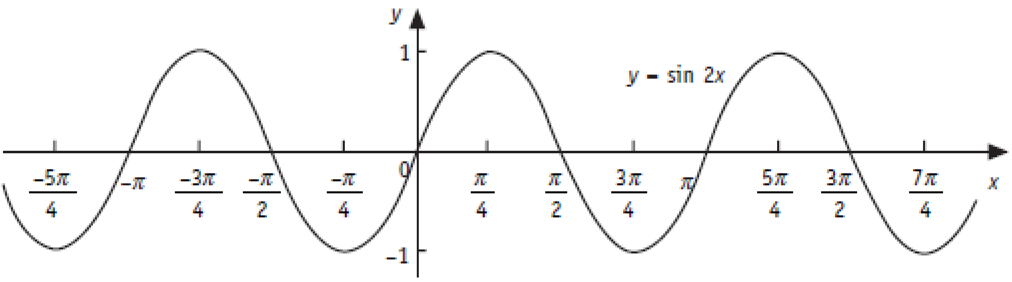
Amplitude = 2   Amplitude = 3   Amplitude = 4

Period = 2π Period = $\dfrac{2\pi}{b}=\dfrac{2\pi}{4}=\dfrac{\pi}{2}$ Period = $\dfrac{2\pi}{b}=\dfrac{2\pi}{\pi}=2$

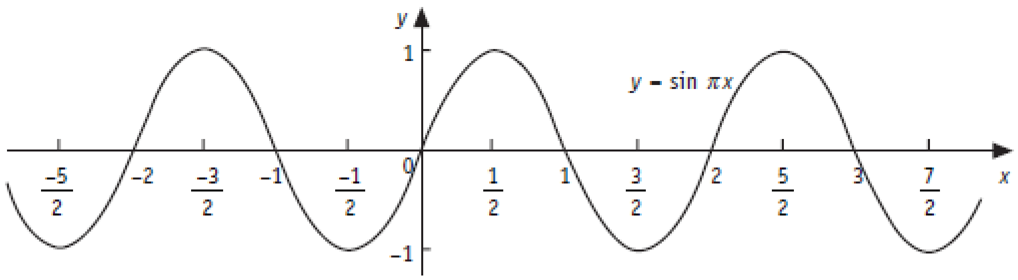
6. (i) $y=2\sin x$



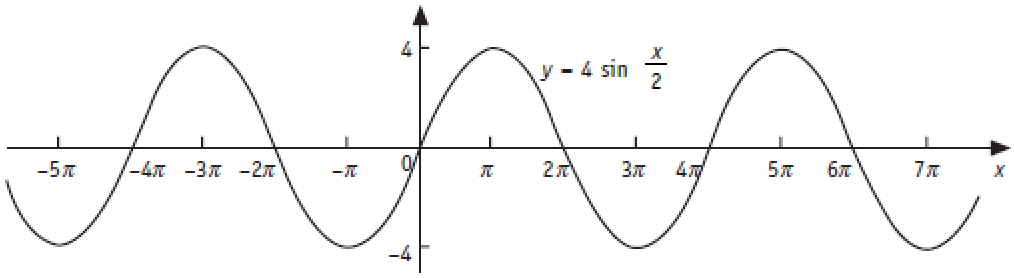
(ii) $y=\sin 2x$



(iii) $y = \sin \pi x$



(iv) $y = 4\sin\dfrac{x}{2}$



7. (i) $x: x \neq \pm\dfrac{\pi}{2}, \pm\dfrac{3\pi}{2}, \pm\dfrac{5\pi}{2}, \pm\dfrac{7\pi}{2}, \ldots$ (ii) all real y $\{y \in \mathbb{R}\}$ (iii) π (iv) $\dfrac{\pi}{n}$

8. (i) $y = \tan 4x$

No amplitude

Period = $\dfrac{\pi}{4}$

(ii) $y = 3\tan x$

No amplitude

Period = π

(iii) $y = -\tan \pi x$

No amplitude

Period = 1

(iv) $y = 2\tan\dfrac{x}{2}$

No amplitude

Period = 2π

(v) $y = \tan 2x$

No amplitude

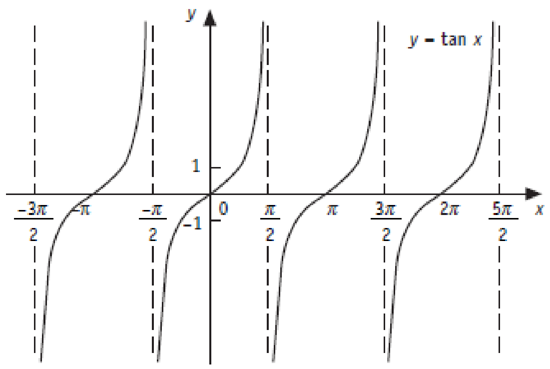
Period = $\dfrac{\pi}{b} = \dfrac{\pi}{2}$

(vi) $y = 1 - 4\tan 3x$

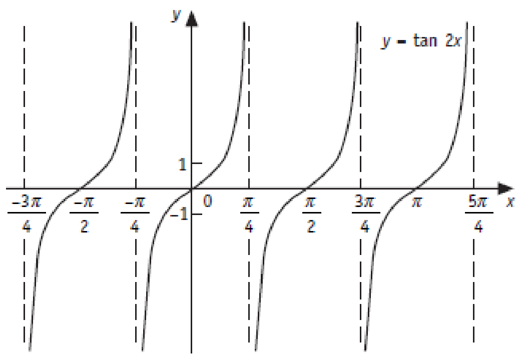
No amplitude

Period = $\dfrac{\pi}{b} = \dfrac{\pi}{3}$

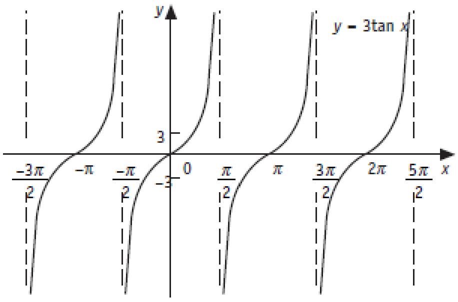
9. (i) $y = \tan x$



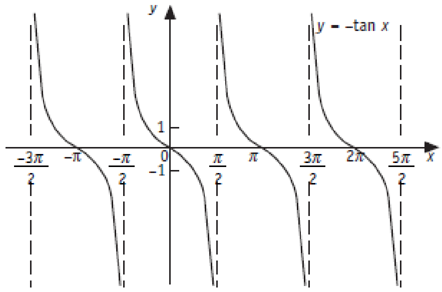
(ii) $y = \tan 2x$



(iii) $y = 3\tan x$



(iv) $y = -\tan x$



10. (i) $y = 10\sin x$

No. cycles = b = 1

Period = $\dfrac{2\pi}{1} = 2\pi$

(ii) $y = 1 + 8\sin 4x$

No. cycles = b = 4

Period = $\dfrac{2\pi}{4} = 0.5\pi$

(iii) $y = 1 - \sin\dfrac{x}{2}$

No. cycles = b = $\dfrac{1}{2}$

Period = $\dfrac{2\pi}{\frac{1}{2}} = 4\pi$

(iv) $y = -2\cos\left(\dfrac{x}{3}\right) + 2$

No. cycles = b = $\dfrac{1}{3}$

Period = $\dfrac{2\pi}{\frac{1}{3}} = 6\pi$

11. (i) Amplitude = 2 There are 4 cycles in 2π $\therefore b = 3$

$\therefore$ Equation is $y = 2\sin 3x$

(ii) Amplitude = 1 There are 2 cycles in 2π $\therefore b = 2$

$\therefore$ Equation is $y = \sin 2x$

(iii) Amplitude = 1 There is 1 cycle in 2π $\therefore b = 1$

Graph is shifted up by 1 unit

$\therefore$ Equation is $y = 1 + \sin x$

12.
$$Period = \frac{2\pi}{b}$$
$$4\pi = \frac{2\pi}{b}$$
$$b = \frac{2\pi}{4\pi} = \frac{1}{2}$$
$$y = 3\sin\left(\frac{x}{2}\right) - 2$$

13.
$$Period = \frac{2\pi}{b}$$
$$\pi = \frac{2\pi}{b}$$
$$b = \frac{2\pi}{\pi} = 2$$
$$y = 4\sin(2x) + 3$$

14. $g(x) = 1 + 2\sin\left(x + \dfrac{\pi}{6}\right)$